

CSC311H5 Introduction to Machine Learning

Lecture 8

University of Toronto Mississauga

Agenda

Until last week, we have been studying **discriminative classifiers** for supervised classification.

- A discriminative classifier learns a mapping directly from the space of inputs $\mathbf{x}$ to class labels $c \in \{0, 1, 2, \dots, K\}$

Today: **Generative model**

- Try to model $p(\mathbf{x}, c)$. If we know $p(c)$ we can easily compute $p(\mathbf{x}|c)$.
- Model $p(\mathbf{x}|c)$, and then apply Bayes Rule to derive $p(c|\mathbf{x})$.

Section 1

Generative Classifiers

Generative vs Discriminative Approaches to Classification

Discriminative approach: estimate parameters of decision boundary/class separator directly from labeled examples.

- Model $p(c|\mathbf{x})$ directly (logistic regression models)
 - ▶ Use the symbol c to denote the target class t
- Learn mappings from inputs to classes (linear/logistic regression, decision trees etc)
- Tries to solve: How do I separate the classes?

Generative approach: model the distribution of inputs characteristic of the class (Bayes classifier).

- Model $p(\mathbf{x}|c)$
- Apply Bayes Rule to derive $p(c|\mathbf{x})$.
- Tries to solve: What does each class “look” like?

Key difference: is there a distributional assumption over inputs?

Generative Classifiers Examples

Suppose that we have the following data set:

x1 (ChestPain)	x2 (Cholesterol)	...	c (Heart Disease)
...	...	...	...

Here is a *model* of how data is generated:

- Sample c from its prior distribution $p(c)$, which is a Bernoulli distribution
- Sample $x_1, x_2, \dots, x_D$ from the distribution $p(x_1, x_2, \dots, x_D | c)$
 - ▶ Two separate distributions that each describe what each class “look” like
 - ★ $p(x_1, x_2, \dots, x_D | c = 1)$
 - ★ $p(x_1, x_2, \dots, x_D | c = 0)$

Inference in a Generative Classifier

If we have a new data patient with features $x_1, x_2, \dots, x_D$, how do we predict whether this patient has heart disease? That is, how do we compute

$$p(c = 1 | x_1, x_2, \dots, x_D)$$

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Bayes Rule!

$$p(c = 1 | x_1, x_2, \dots, x_D) = \frac{p(x_1, x_2, \dots, x_D | c = 1)p(c = 1)}{p(x_1, x_2, \dots, x_D)}$$

Learning in a Generative Classifier

Learning requires learning the parameters for the distributions that describe what each class “look” like

- $p(x_1, x_2, \dots, x_D | c = 1)$
- $p(x_1, x_2, \dots, x_D | c = 0)$

We can learn the parameters of these distributions, e.g. via MLE

Example: Movie Review Classification (Lab 8)

Examples of Positive Reviews:

- he was both funny and sad the script worked well too
- i enjoyed this movie as a kid when it came out and to this day still do i am surprised not too many people know about it cool flick
- but the plot i heard was so great was so predictable plus the only cool character gets killed off in the middle of the movie wrong in short not much happens

Examples of Negative Reviews:

- for a comedy this movie simply isnt good or funny enough
- the direction is terrible the acting is so so rest of the cast are average at best overall not worth your time or money
- all the way though i was thinking to myself oh god why why does going into the house make her come after you it doesnt make sense you can see everything coming which just left you feeling that there was no point in watching oh shes behind her

Discriminative classifiers for Movie Reviews

Learn mappings directly from the space of inputs $\mathbf{x}$ to class labels $c \in \{0, 1, 2, \dots, K\}$

Feature $\mathbf{x} \rightarrow$ Class Probability $p(c|\mathbf{x})$

"he was both funny and sad the script ..." $\rightarrow$ "Positive"

"for a comedy this movie simply isnt ..." $\rightarrow$ "Negative"

Generative classifiers for Movie Reviews

Try to model $p(\mathbf{x}, c)$. If we know $p(c)$ we can easily compute $p(\mathbf{x}|c)$.

We perform classification via **Bayes rule**

Probability of feature given label $p(\mathbf{x}|c) \leftarrow$ Class Label c

"he was both funny and sad the script ..." $\leftarrow$ "Positive"

"for a comedy this movie simply isnt ..." $\leftarrow$ "Negative"

Obtaining Binary Features of Text

Binary Bag-of-words Features: represent each review as a binary vector $\mathbf{x}$, where each $x_i \in \{0, 1\}$ corresponding to whether word i is present/absent in a review.

- Vocabulary:
 - ▶ “a”: 0
 - ▶ “and”: 1
 - ▶ ...
 - ▶ “fun”: 0
 - ▶ “funny”: 1
 - ▶ “future”: 0
 - ▶ ...
 - ▶ “was”: 1

For the lab, the reviews are curated that we have 1000 words in our vocabulary.

Exercise: Binary Bag-of-words Features

Exercise: Write down the binary bag-of-words features for the following reviews. Assume that the vocabulary consists of only the words appearing in these reviews.

- he was funny and sad
- movie was sad
- great movie
- movie was not funny
- movie was not sad

Bag-of-words features

and	he	funny	movie	sad	not	was	great
1	1	1	0	1	0	1	0
0	0	0	1	1	0	1	0
0	0	0	1	0	0	0	1
0	0	1	1	0	1	1	0
0	0	0	1	1	1	1	0

Bayes Classifier: Inference

Given features $\mathbf{x} = [x_1, x_2, \dots, x_D]^T$ we want to compute class probabilities using Bayes Rule:

$$\underbrace{p(c|\mathbf{x})}_{\text{Pr. class given words}} = \frac{p(\mathbf{x}, c)}{p(\mathbf{x})} = \frac{\overbrace{p(\mathbf{x}|c})^{\text{Pr. words given class}} p(c)}{p(\mathbf{x})}$$

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In other words:

$$\text{Posterior} = \frac{\text{Class likelihood} \times \text{Class Prior}}{\text{Evidence}}$$

Bayes Classifier: Evidence

Q: How can we compute $p(\mathbf{x})$ for the two class case?

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$$\begin{aligned} p(\mathbf{x}) &= \sum_c p(\mathbf{x}|c)p(c) \\ &= p(\mathbf{x}|c=0)p(c=0) + p(\mathbf{x}|c=1)p(c=1) \end{aligned}$$

Q: Do we need to compute $p(\mathbf{x})$ if all we want is a binary prediction?

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Q: Do we need to compute $p(\mathbf{x})$ if all we want is a binary prediction?

No! If all we want is to compare $p(c=1|\mathbf{x})$ and $p(c=0|\mathbf{x})$, we can compare their respective numerators instead!

$$\begin{aligned} p(c=0|\mathbf{x}) &\propto p(\mathbf{x}|c=0)p(c=0) \\ p(c=1|\mathbf{x}) &\propto p(\mathbf{x}|c=1)p(c=1) \end{aligned}$$

Towards Naive Bayes

- Assume we have two classes: positive and negative. We have a dictionary of D words, and binary features $\mathbf{x} = [x_1, \dots, x_D]$ saying whether each word appears in the review.

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- If we define a joint distribution $p(c, x_1, \dots, x_D)$, this gives enough information to determine $p(c)$ and $p(\mathbf{x}|c)$.
- Problem: specifying a joint distribution over $D + 1$ binary variables requires $2^{D+1} - 1$ entries. This is computationally prohibitive and would require an absurd amount of data to fit.

Towards Naive Bayes

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- If we define a joint distribution $p(c, x_1, \dots, x_D)$, this gives enough information to determine $p(c)$ and $p(\mathbf{x}|c)$.
- Problem: specifying a joint distribution over $D + 1$ binary variables requires $2^{D+1} - 1$ entries. This is computationally prohibitive and would require an absurd amount of data to fit.
- We will impose **structure** on the distribution such that:
 - ▶ it can be **compactly** represented
 - ▶ **learning** and **inference** are both tractable

Section 2

The Naive Bayes Classifier

Naive Bayes

A **Naive Bayes** classifier assumes that the word features x_i are **conditionally independent** given the class c . (the naive assumption)

$$p(c, x_1, x_2, \dots, x_D) = p(c)p(x_1|c)p(x_2|c) \cdots p(x_D|c)$$

Here, x_i and x_j are independent under the conditional distribution $p(\mathbf{x}|c)$.
(Note: this does *not* mean that x_i and x_j are independent!)

Interpretation example: “knowing that a review is positive, the knowledge of whether the word ‘great’ appears in the review does not tell us whether the word ‘funny’ also appears”

Naive Bayes Parameters

$$p(c, x_1, x_2, \dots, x_D) = p(c)p(x_1|c)p(x_2|c) \cdots p(x_D|c)$$

When x_j and c are binary, we have a compact representation of the joint distribution as a product of Bernoulli distributions:

- $P(c = 1) = \pi$
 - ▶ the class prior
- $P(x_j = 1|c) = \theta_{jc}$
 - ▶ probability of word j appearing in the review, given class c (e.g., word “fun” appearing in positive review)

Q: How many parameters do we have in this model, in total?

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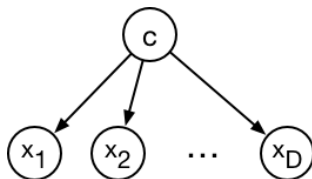
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Q: How many parameters do we have in this model, in total?

- For $P(c = 1)$: 1 parameter
- For $P(x_j = 1|c)$: D parameters per class $\times$ 2 classes
- So $2D + 1$ parameters total (compared to $2^{D+1} - 1$ before)

Bayes Nets

Naive Bayes is a simple example of a **directed graphical model**, or **Bayesian network**



- This graph structure means the joint distribution factorizes as a product of conditional distributions for each variable given its parent(s).
- Intuitively, you can think of the edges as reflecting a causal structure. But mathematically, this doesn't hold without additional assumptions.

Naive Bayes Log Likelihood

The parameters can be learned efficiently because the log-likelihood decomposes into independent terms for each feature.

$$\ell(\theta) = \sum_{i=1}^N \log p(c^{(i)}, \mathbf{x}^{(i)}) = \sum_{i=1}^N \log \left\{ p(\mathbf{x}^{(i)} | c^{(i)}) p(c^{(i)}) \right\}$$

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Naive Bayes Learning

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Each of these log-likelihood terms depends on different sets of parameters, so they can be optimized independently:

- Optimize $\sum_{i=1}^N \log p(c^{(i)})$, the log probability of observing the (positive/negative) labels in our data set
 - ▶ $p(c^{(i)})$ is a Bernoulli distribution

Naive Bayes Learning

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Each of these log-likelihood terms depends on different sets of parameters, so they can be optimized independently:

- Optimize $\sum_{i=1}^N \log p(c^{(i)})$, the log probability of observing the (positive/negative) labels in our data set
 - ▶ $p(c^{(i)})$ is a Bernoulli distribution
- Optimize $\sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)})$, which can be further decomposed so we learn each θ_{jc} separately
 - ▶ Each $p(x_j^{(i)} | c^{(i)})$ is also a Bernoulli distribution

Naive Bayes: Learning the Prior

For the prior we maximize: $\sum_{i=1}^N \log p(c^{(i)})$

Let $p(c^{(i)})$ be the Bernoulli distribution, with $p(c^{(i)}) = \pi$ (Here π is a parameter to be optimized.)

$$p(c^{(i)}) = \pi^{c^{(i)}} (1 - \pi)^{1-c^{(i)}}$$

We wish to optimize the log-likelihood:

$$\sum_{i=1}^N \log p(c^{(i)}) = \sum_{i=1}^N c^{(i)} \log \pi + \sum_{i=1}^N (1 - c^{(i)}) \log(1 - \pi)$$

We obtain MLEs by setting derivatives to zero:

$$\hat{\pi} = \frac{\sum_i \mathbb{I}[c^{(i)} = 1]}{N} = \frac{\# \text{ positive reviews}}{\text{total } \# \text{ samples}}$$

Exercise: Learning the Prior

Exercise: Compute the MLE estimate for $P(c = 1) = \pi$ for this data set:

- he was funny and sad, “positive”
- movie was sad, “positive”
- great movie, “positive”
- movie was not funny, “negative”
- movie was not sad, “negative”

Exercise: Learning the Prior

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$$\hat{\pi} = \frac{3}{5}$$

Naive Bayes: Learning each θ_{jc}

To maximize $\sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)})$, we separate the classes:

$$\begin{aligned} \sum_{i=1}^N \log p(x_j^{(i)} | c^{(i)}) &= \sum_{i=1}^N c^{(i)} \left\{ x_j^{(i)} \log \theta_{j1} + (1 - x_j^{(i)}) \log(1 - \theta_{j1}) \right\} \\ &\quad + \sum_{i=1}^N (1 - c^{(i)}) \left\{ x_j^{(i)} \log \theta_{j0} + (1 - x_j^{(i)}) \log(1 - \theta_{j0}) \right\} \end{aligned}$$

- Here, we have let $\theta_{jc} = p(x_j^{(i)} = 1 | c)$, Bernoulli distribution
- So that $p(x_j^{(i)} | c) = \theta_{jc}^{x_j^{(i)}} (1 - \theta_{jc})^{1-x_j^{(i)}}$

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- Here, we have let $\theta_{jc} = p(x_j^{(i)} = 1 | c)$, Bernoulli distribution
- So that $p(x_j^{(i)} | c) = \theta_{jc}^{x_j^{(i)}} (1 - \theta_{jc})^{1-x_j^{(i)}}$

We obtain MLEs by setting derivatives to zero, for example:

$$\hat{\theta}_{jc} = \frac{\sum_i \mathbb{I}[x_j^{(i)} = 1 \ \& \ c^{(i)} = c]}{\sum_i \mathbb{I}[c^{(i)} = c]} \quad \text{for } \underline{c=1} \quad \frac{\text{\#word } j \text{ appears in pos reviews}}{\text{\# positive reviews}}$$

Exercise: Learning each θ_{jc}

Exercise: Compute the MLE estimates for each θ_{jc} in this data set:

- he was funny and sad, “positive”
- movie was sad, “positive”
- great movie, “positive”
- movie was not funny, “negative”
- movie was not sad, “negative”

Example: For the word j = “movie”

- $\theta_{movie,1} = \frac{2}{3}$
- $\theta_{movie,0} = \frac{2}{2}$

Learning each θ_{jc}

word	“positive”	“negative”
movie	2/3	2/2
sad	2/3	1/2
he	1/3	0/2
was	2/3	2/2
not	0/3	2/2
funny	1/3	1/2
great	1/3	0/2
and	1/3	0/2

Naive Bayes Inference

- We predict the category by performing **inference** in the model.
- Apply **Bayes' Rule**:

$$p(c|\mathbf{x}) = \frac{p(c)p(\mathbf{x}|c)}{\sum_{c'} p(c')p(\mathbf{x}|c')} = \frac{p(c) \prod_{j=1}^D p(x_j|c)}{\sum_{c'} p(c') \prod_{j=1}^D p(x_j|c')}$$

- We need not compute the denominator if we're simply trying to determine the most likely c .
- Shorthand notation:

$$p(c|\mathbf{x}) \propto p(c) \prod_{j=1}^D p(x_j|c)$$

- For input $\mathbf{x}$, predict by comparing the values of $p(c) \prod_{j=1}^D p(x_j|c)$ for different c (e.g. choose the largest).

Exercise: Naive Bayes Inference

Exercise: Using the parameters computed earlier, estimate the probability that this review is positive.

- movie was funny

Exercise: What about this review?

- he was not funny

Exercise: Naive Bayes Inference (Part 1)

Exercise: Using the parameters computed earlier, estimate the probability that the review “movie was funny” is positive.

Exercise: Naive Bayes Inference (Part 1)

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$$\begin{aligned}P(c = 1|\mathbf{x}) &\propto P(x_{and} = 0|c = 1)P(x_{he} = 0|c = 1)P(x_{funny} = 1|c = 1) \\&\quad P(x_{movie} = 1|c = 1)P(x_{sad} = 0|c = 1)P(x_{not} = 0|c = 1) \\&\quad P(x_{was} = 1|c = 1)P(x_{great} = 0|c = 1)P(c = 1) \\&= (1 - \theta_{and,1})(1 - \theta_{he,1})\theta_{funny,1} \\&\quad \theta_{movie,1}(1 - \theta_{sad,1})(1 - \theta_{not,1}) \\&\quad \theta_{was,1}(1 - \theta_{great,1})\pi \\&= (1 - \frac{1}{3})(1 - \frac{1}{3})\frac{1}{3}\frac{2}{3}(1 - \frac{2}{3})(1 - \frac{0}{3})\frac{1}{3}(1 - \frac{1}{3})\frac{3}{5}\end{aligned}$$

Exercise: Naive Bayes Inference (Part 1)

Exercise: Using the parameters computed earlier, estimate the probability that the review “movie was funny” is positive.

$$\begin{aligned}P(c = 1|\mathbf{x}) &\propto P(x_{and} = 0|c = 1)P(x_{he} = 0|c = 1)P(x_{funny} = 1|c = 1) \\&\quad P(x_{movie} = 1|c = 1)P(x_{sad} = 0|c = 1)P(x_{not} = 0|c = 1) \\&\quad P(x_{was} = 1|c = 1)P(x_{great} = 0|c = 1)P(c = 1) \\&= (1 - \theta_{and,1})(1 - \theta_{he,1})\theta_{funny,1} \\&\quad \theta_{movie,1}(1 - \theta_{sad,1})(1 - \theta_{not,1}) \\&\quad \theta_{was,1}(1 - \theta_{great,1})\pi \\&= (1 - \frac{1}{3})(1 - \frac{1}{3})\frac{1}{3}\frac{2}{3}(1 - \frac{2}{3})(1 - \frac{0}{3})\frac{1}{3}(1 - \frac{1}{3})\frac{3}{5}\end{aligned}$$

... then perform the same computation for the numerator of $P(c = 0|\mathbf{x})$.
Then compute $P(\mathbf{x})$, and obtain the desired probability.

Exercise: Naive Bayes Inference (Part 2)

Exercise: Using the parameters computed earlier, estimate the probability that the review “he was not funny” is positive.

Exercise: Naive Bayes Inference (Part 2)

Exercise: Using the parameters computed earlier, estimate the probability that the review “he was not funny” is positive.

$$\begin{aligned}P(c = 1|\mathbf{x}) &\propto P(x_{not} = 1|c = 1) \times P(\dots) \times \dots \\&= \theta_{not,1} \times \dots \\&= 0 \times \dots \\&= 0\end{aligned}$$

$$\begin{aligned}P(c = 0|\mathbf{x}) &\propto P(x_{movie} = 0|c = 1) \times P(\dots) \times \dots \\&= (1 - \theta_{movie,0}) \times \dots \\&= (1 - 1) \times \dots \\&= 0\end{aligned}$$

This is a problem!

Naive Bayes: Summary so Far

- Naive Bayes is an amazingly cheap learning algorithm!
- **Training time:** estimate parameters using maximum likelihood
 - ▶ Compute co-occurrence counts of each feature with the labels.
 - ▶ Requires only one pass through the data!
- **Test time:** apply Bayes' Rule
 - ▶ Cheap because of the model structure. (For more general models, Bayesian inference can be very expensive and/or complicated.)
- We covered the Bernoulli case for simplicity. But our analysis easily extends to other probability distributions.
- Unfortunately, it's usually less accurate in practice compared to discriminative models due to its "naive" independence assumption.

But how do we deal with the issue we saw in the last slide?

Section 3

Bayesian Parameter Estimation

MLE issue: Data Sparsity

Maximum likelihood has a pitfall: if you have too little data, it can overfit.

E.g., Suppose we flip a (possibly biased) coin 2 times, and get tails both times. What is our estimate of the probability of getting heads?

$$\theta_{\text{ML}} = \frac{N_H}{N_T + N_H} = 0$$

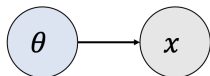
where N_H is the number of heads observed, N_T is the number of tails observed.

Because we never observed a “heads” flip, we assign this outcome probability 0.

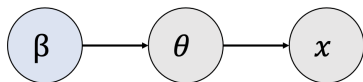
This problem is known as **data sparsity**.

Bayesian Parameter Estimation

In maximum likelihood, the observations are treated as random variables, but the parameters are not.



The **Bayesian** approach treats the parameters as random variables as well. β is the set of parameters in the prior distribution of θ .



- To define a Bayesian model, we need to specify two distributions:
 - ▶ The **prior distribution** $p(\theta)$, which encodes our beliefs about the parameters *before* we observe the data
 - ▶ The **likelihood** $p(\mathcal{D}|\theta)$, same as in maximum likelihood

Bayesian Parameter Estimation

When we **update** our beliefs about θ based on the observations, we compute the **posterior distribution** of θ using Bayes' Rule:

$$p(\theta|\mathcal{D}) = \frac{p(\theta)p(\mathcal{D}|\theta)}{\int p(\theta')p(\mathcal{D}|\theta') d\theta'}$$

We rarely ever compute the denominator explicitly. In general, it is computationally intractable.

Bayesian Parameter Estimation for the Coin Example

For the coin example, we already know the likelihood:

$$\mathcal{L}(\theta) = p(\mathcal{D}|\theta) = \theta^{N_H}(1 - \theta)^{N_T}$$

It remains to specify the prior $p(\theta)$.

- We can choose an **uninformative prior**, which assumes as little as possible.
- A reasonable choice is the **uniform prior**: choose $p(\theta)$ to be a uniform distribution
- In this case, the posterior is proportional to the likelihood

Bayesian Parameter Estimation: Beta Distribution Prior

But for the coin example, our experience tells us that θ being 0.5 is more likely than θ being 0.99

- One particularly useful prior that lets us specify this is the **beta distribution**:

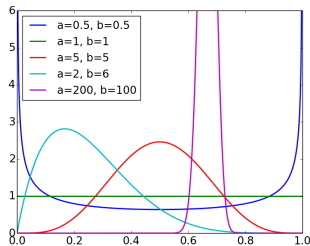
$$p(\theta; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1} (1-\theta)^{b-1}$$

- This notation for proportionality lets us ignore the normalization constant:

$$p(\theta; a, b) \propto \theta^{a-1} (1-\theta)^{b-1}$$

The Beta Distribution

Beta distribution for various values of a , b :



Some observations:

- The expectation $\mathbb{E}[\theta] = a/(a + b)$
- The distribution gets more peaky when a and b are large.
- The uniform distribution is the special case where $a = b = 1$.

The beta distribution is used for a prior for the Bernoulli distribution.

Bayesian Parameter Estimation: Computing the Posterior

Computing the posterior distribution:

$$\begin{aligned} p(\theta|\mathcal{D}) &\propto p(\theta)p(\mathcal{D}|\theta) \\ &\propto \left[\theta^{a-1}(1-\theta)^{b-1} \right] \left[\theta^{N_H}(1-\theta)^{N_T} \right] \\ &= \theta^{a-1+N_H}(1-\theta)^{b-1+N_T}. \end{aligned}$$

This is just a beta distribution with parameters $N_H + a$ and $N_T + b$

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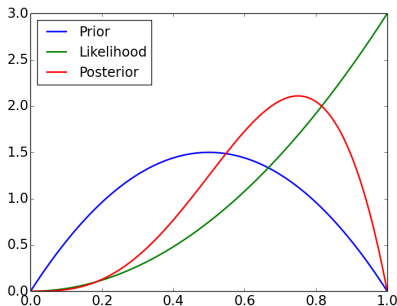
- The posterior expectation of θ is:

$$\mathbb{E}[\theta|\mathcal{D}] = \frac{N_H + a}{N_H + N_T + a + b}$$

- The parameters a and b of the prior can be thought of as **pseudo-counts**. The reason this works is that the prior and likelihood have the same functional form. This phenomenon is known as **conjugacy** (conjugate priors)

Bayesian Inference for Coin Flip (Small Data)

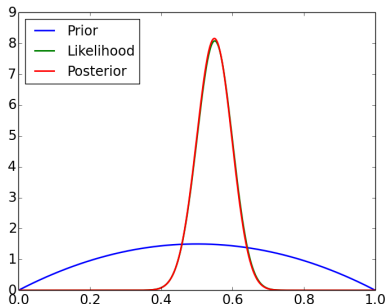
Small data setting $N_H = 2, N_T = 0$



When we don't have enough observations, the prior significantly affects the posterior estimate.

Bayesian Inference for Coin Flip (Large Data)

Large data setting $N_H = 55, N_T = 55$



When we have enough observations, the **data overwhelm the prior**.
(Posterior & Likelihood overlaps!)

What do we actually do with the posterior?

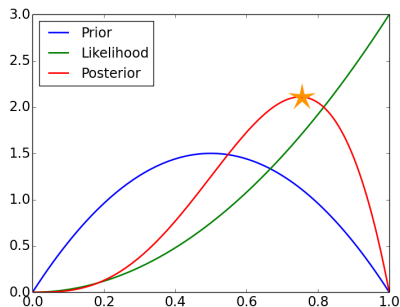
The **posterior predictive distribution** is the distribution over future observables given the past observations. We compute this by marginalizing out the parameter(s):

$$p(\mathcal{D}'|\mathcal{D}) = \int p(\theta|\mathcal{D})p(\mathcal{D}'|\theta) d\theta$$

Can be costly to estimate this quantity.

Maximum A-Posteriori Estimation

Maximum A-Posteriori Estimation: find the most likely parameter settings under the posterior



Maximum A-Posteriori Estimation

Using MAP converts the Bayesian parameter estimation problem into a maximization problem

$$\begin{aligned}\hat{\theta}_{MAP} &= \arg \max_{\theta} p(\theta|\mathcal{D}) \\ &= \arg \max_{\theta} p(\theta, \mathcal{D}) \\ &= \arg \max_{\theta} p(\theta) p(\mathcal{D}|\theta) \\ &= \arg \max_{\theta} \log p(\theta) + \log p(\mathcal{D}|\theta)\end{aligned}$$

Maximum A-Posteriori Estimation for Coin Flip

In the coin flip example, we have the following joint probability:

$$\begin{aligned}\log p(\theta, \mathcal{D}) &= \log p(\theta) + \log p(\mathcal{D}|\theta) \\ &= \text{Const} + (a - 1) \log \theta + (b - 1) \log(1 - \theta) \\ &\quad + N_H \log \theta + N_T \log(1 - \theta) \\ &= \text{Const} + (N_H + a - 1) \log \theta + (N_T + b - 1) \log(1 - \theta)\end{aligned}$$

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Maximize by finding a critical point

$$0 = \frac{d}{d\theta} \log p(\theta, \mathcal{D}) = \frac{N_H + a - 1}{\theta} - \frac{N_T + b - 1}{1 - \theta}$$

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Solving for θ :

$$\hat{\theta}_{MAP} = \frac{N_H + a - 1}{N_H + N_T + a + b - 2}$$

Comparison of Estimates in the Coin Flip Example

	Formula	$N_H = 2, N_T = 0$	$N_H = 55, N_T = 45$
$\hat{\theta}_{ML}$	$\frac{N_H}{N_H + N_T}$	1	$\frac{55}{100} = 0.55$
$\mathbb{E}[\theta \mathcal{D}]$	$\frac{N_H + a}{N_H + N_T + a + b}$	$\frac{4}{6} \approx 0.67$	$\frac{57}{104} \approx 0.548$
$\hat{\theta}_{MAP}$	$\frac{N_H + a - 1}{N_H + N_T + a + b - 2}$	$\frac{3}{4} = 0.75$	$\frac{56}{102} \approx 0.549$

$\hat{\theta}_{MAP}$ assigns nonzero probabilities as long as $a, b > 1$.

Exercise: Learning each θ_{jc}

Exercise Compute the MAP estimate for π and each θ_{jc} in this data set, assuming a Beta distribution priors for each parameter with $a = 2$, $b = 2$.

- he was funny and sad, “positive”
- movie was sad, “positive”
- great movie, “positive”
- movie was not funny, “negative”
- movie was not sad, “negative”

Then, use your parameters to estimate the probabilities that these reviews are positive:

- movie was funny
- he was not funny